

PROJECT DESIGN DOCUMENT

# MLE Package WWTP

## 2,000 GPD · Fixed PVA IFAS · CRS Nanobubble Aeration

**Effluent Target: "d10 mg/L NO *f*-N**

Site: Decentralized package site · Generic, US

Document type: Preliminary / Owner Budget Package

Date: July 10, 2026

Process: MLE + Fixed PVA IFAS + CRS NBG + NH<sub>3</sub>/NO<sub>3</sub>/TSS + AI optimization

Prepared for: Design, permitting, vendor RFQ, and MooreVIEW SCADA integration

Estimates are planning-level (±25–35%) until site-specific engineering and vendor quotes are obtained.

# 1. Executive Summary

This document defines a wastewater treatment plant (WWTP) for Decentralized package site (Generic, US), rated for 1.4 GPM (2,000 GPD average flow). The biological process is Modified Ludzack-Ettinger (MLE) biological nitrogen removal configured as hybrid integrated fixed-film activated sludge (IFAS) with fixed PVA foam carriers, CRS (Chemical Reduction Systems) nanobubble generators, s::can recirculating optical measurements (BOD, NH f, NO f, TSS, pH, ORP per phase), and AI-based process optimization.

Configuration: 4,000 gal Package Reactor. The design target is "d10 mg/L nitrate-nitrogen (NO f-N) in the final effluent, with ammonia-nitrogen (NH f-N) <1 mg/L and secondary quality BOD .../TSS "d10 mg/L.

| Metric                                  | Value               |
|-----------------------------------------|---------------------|
| Design average flow                     | 2,000 GPD (1.4 gpm) |
| Design peak flow                        | 4,000 GPD           |
| Biological volume                       | 4,000 gal           |
| Footprint (approx.)                     | 120-150 ft²         |
| Mid-tier installed CAPEX                | ~\$200,000          |
| Total project budget (with contingency) | ~\$275,000          |
| Annual OPEX (mid)                       | \$6,000-9,000       |

# 2. Design Basis

| Parameter                  | Value               | Notes                                          |
|----------------------------|---------------------|------------------------------------------------|
| Average flow (Q)           | 1.4 GPM (2,000 GPD) | 1.4 gpm design                                 |
| Peak flow                  | 4,000 GPD           | Diurnal / seasonal peaking                     |
| Influent BOD ...           | 200 mg/L            | Carbon for denitrification                     |
| Influent BOD ...           | 200 mg/L            | s::can INF_BOD · C:N ratio for denitrification |
| Influent TSS               | 180 mg/L            | s::can INF_TSS · AI baseline                   |
| Influent TKN               | 40 mg/L             | Typical municipal strength                     |
| Influent NH f-N            | 30 mg/L             |                                                |
| Influent total nitrogen    | 45 mg/L             | Includes organic N                             |
| Design minimum temperature | 10°C                | Winter nitrification driver                    |
| Effluent NO f-N limit      | "d10 mg/L           | Monthly average permit target                  |
| Effluent NH f-N            | <1 mg/L             | Design objective                               |
| Effluent BOD ... / TSS     | "d10 / "d10 mg/L    | Secondary treatment standard                   |

# 3. Process Description

Wastewater passes through preliminary screening (if required by local code), then enters the pre-anoxic zone of the MLE reactor. Return activated sludge (RAS) is mixed with influent in the anoxic zone where denitrification occurs using influent readily biodegradable COD (rbCOD). Mixed liquor underflows to the aerobic IFAS zone where autotrophic nitrifiers oxidize ammonia on both suspended floc and fixed PVA biofilm. Internal recycle (IR) pumps return nitrified mixed liquor from the aerobic zone to the anoxic zone.

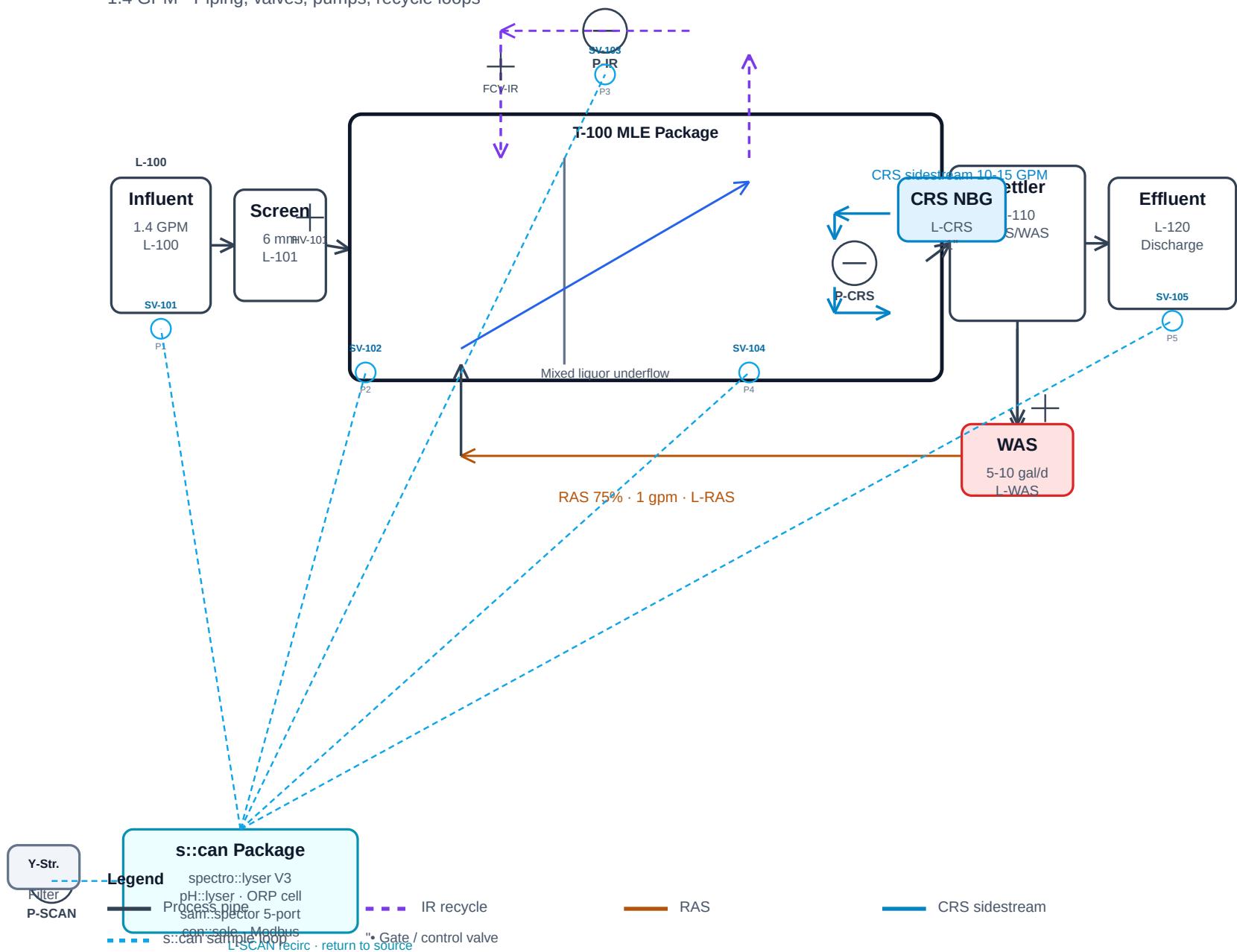
Oxygen is supplied primarily through a continuous sidestream loop: mixed liquor is pumped through a CRS Nanobubble Generator (NBG) and returned to the aerobic zone. A small fine-bubble diffuser and trim blower maintains dissolved oxygen (DO) setpoint during peak load or cold weather. Nano-bubbles are not applied in the anoxic zone.

Online measurements use a recirculating scan optical analyzer package (spectrolyser, pHlyser, ORP in flow cell, sam: spector sequencer) that cycles sample ports at each MLE phase: influent (BOD, COD, TSS,  $\text{NH}_4$ , pH, UV254), pre-anoxic ( $\text{NO}_x$ , ORP, pH, TSS, residual COD), IR stream, aerobic outlet, and final effluent. Influent BOD enables carbon-limited denitrification detection alongside  $\text{NH}_4$ / $\text{NO}_x$ /TSS. Aerobic DO remains on dedicated continuous probes for blower PID. All phase tags historize in MooreVIEW; effluent  $\text{NO}_x$ -N drives IR VFD trim; AI optimizer uses INF\_BOD, C:N ratio, and phase pH/ORP within operator bounds.

Treated mixed liquor flows to an integrated lamella/tube settler. Clarified effluent is discharged; RAS returns to anoxic; waste activated sludge (WAS) is removed on an SRT schedule.

# 4. Piping & Instrumentation Diagram (P&ID)

1.4 GPM · Piping, valves, pumps, recycle loops



## 5. Mechanical Drawing

Equipment layout, volumes, materials · 4,000 gal Package Reactor



### Legend

Blue = anoxic zone

Green = aerobic IFAS + PVA

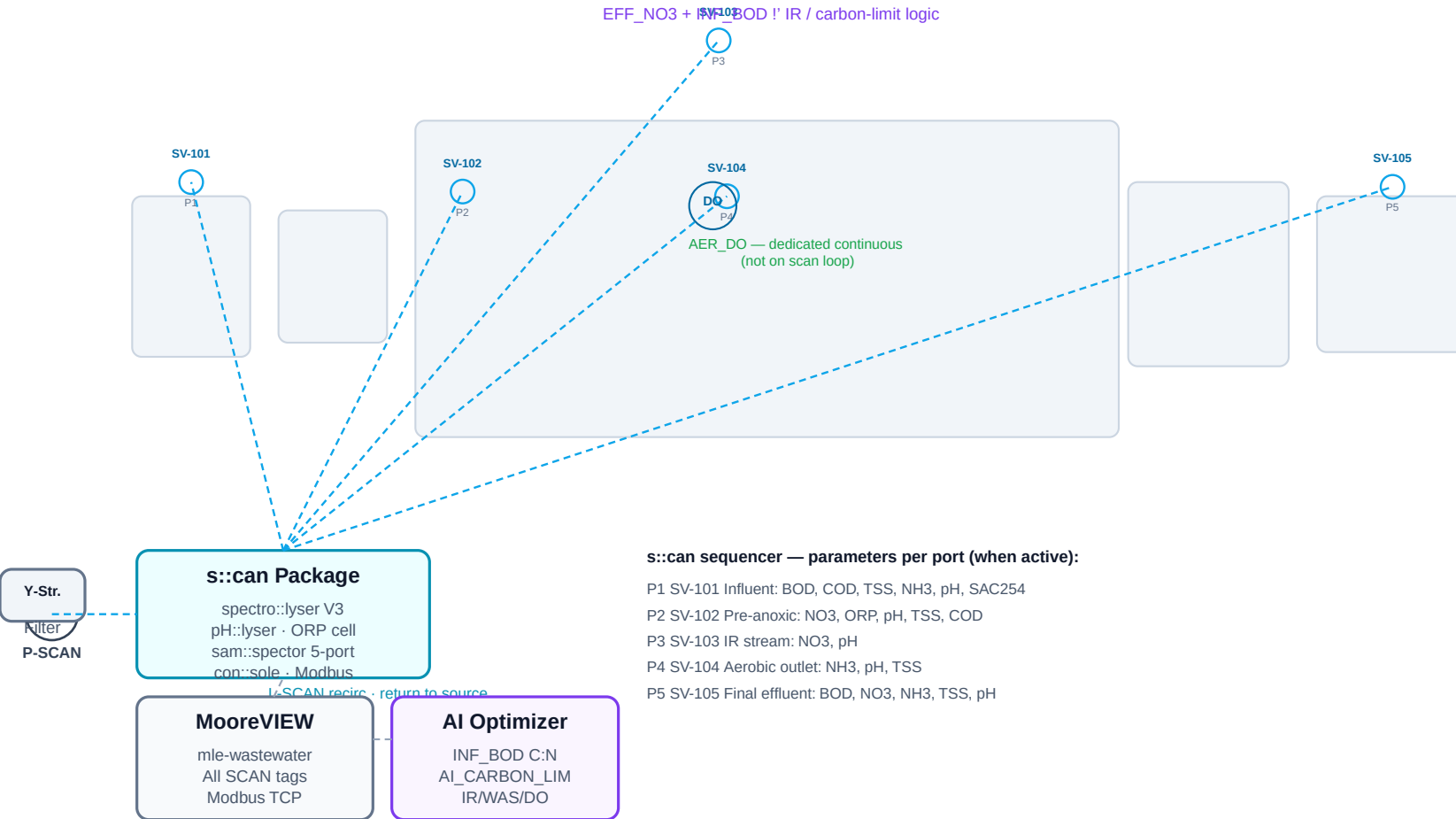
Yellow = clarifier/settler

Cyan = CRS nanobubble loop

Footprint ~120-150 ft<sup>2</sup> · HRT 2 d @ 1.4 GPM

## 6. Instrumentation Diagram

s::can recirculating optical loop · per-phase BOD/pH/ORP · MooreVIEW mle-wastewater



### Legend

■ ■ ■ ■ ■ Sample loop      SV = sequencer sample valve      DO = dedicated aerobic probe      ■ ■ ■ ■ ■ Modbus to SCADA  
 s::can spectro::lyser: BOD/COD/NO3/NH3/TSS/SAC254 · pH::lyser + ORP at anoxic ports · 8 min/port

## 7a. Process Flow Summary

Refer to the drawing set on the preceding landscape pages: Sheet 1 (P&ID) includes the scan recirculating sample loop (L-SCAN, P-SCAN, SV-101...106) to each phase; Sheet 2 (Mechanical) shows the analyzer shelter; Sheet 3 (Instrumentation) lists optical parameters per sequencer port. Main process: influent enters pre-anoxic with RAS; denitrified mixed liquor passes to aerobic IFAS; nitrified liquor is settled and discharged. IR (3.5× Q) returns nitrate-rich liquor to anoxic. CRS sidestream provides nanobubble oxygen in the aerobic zone only.

## 7b. Process Measurement Points (scan Recirculating Loop)

| Tag        | Sample Port               | Parameter | Target / Alarm            | Control Use                                    |
|------------|---------------------------|-----------|---------------------------|------------------------------------------------|
| INF_BOD    | Influent (SV-101)         | BOD       | Track ~200 mg/L · C:N     | Carbon load · C:N ratio · shock detection      |
| INF_COD    | Influent (SV-101)         | COD       | Historian + AI            | Carbon load · C:N ratio · shock detection      |
| INF_TSS    | Influent (SV-101)         | TSS       | Track ~180 mg/L           | Carbon load · C:N ratio · shock detection      |
| INF_NH3    | Influent (SV-101)         | NH3       | Track ~30 mg/L            | Carbon load · C:N ratio · shock detection      |
| INF_PH     | Influent (SV-101)         | pH        | 6.5–8.5                   | Carbon load · C:N ratio · shock detection      |
| INF_SAC254 | Influent (SV-101)         | SAC254    | UV254 organic load        | Carbon load · C:N ratio · shock detection      |
| ANOX_NO3   | Pre-anoxic (SV-102)       | NO3       | <5 mg/L ideal             | Denitrification · MLSS · residual carbon       |
| ANOX_ORP   | Pre-anoxic (SV-102)       | ORP       | -50 to +50 mV anoxic      | Denitrification · MLSS · residual carbon       |
| ANOX_PH    | Pre-anoxic (SV-102)       | pH        | 6.5–8.5                   | Denitrification · MLSS · residual carbon       |
| ANOX_TSS   | Pre-anoxic (SV-102)       | TSS       | 1,500-3,500 mg/L          | Denitrification · MLSS · residual carbon       |
| ANOX_COD   | Pre-anoxic (SV-102)       | COD       | Residual carbon proxy     | Denitrification · MLSS · residual carbon       |
| IR_NO3     | IR stream (SV-103)        | NO3       | 15–30 mg/L typical        | IR trim · nitrate in recycle                   |
| IR_PH      | IR stream (SV-103)        | pH        | 6.5–8.5                   | IR trim · nitrate in recycle                   |
| AER_NH3    | Aerobic outlet (SV-104)   | NH3       | <1 mg/L                   | Nitrification · plus dedicated AER_DO in basin |
| AER_PH     | Aerobic outlet (SV-104)   | pH        | 6.5–8.5                   | Nitrification · plus dedicated AER_DO in basin |
| AER_TSS    | Aerobic outlet (SV-104)   | TSS       | Historian + AI            | Nitrification · plus dedicated AER_DO in basin |
| EFF_BOD    | Final effluent (SV-105)   | BOD       | "d10 mg/L permit          | Permit compliance · IR PID feedback            |
| EFF_NO3    | Final effluent (SV-105)   | NO3       | "d10 mg/L permit · IR PID | Permit compliance · IR PID feedback            |
| EFF_NH3    | Final effluent (SV-105)   | NH3       | <1 mg/L                   | Permit compliance · IR PID feedback            |
| EFF_TSS    | Final effluent (SV-105)   | TSS       | Historian + AI            | Permit compliance · IR PID feedback            |
| EFF_PH     | Final effluent (SV-105)   | pH        | 6.5–8.5                   | Permit compliance · IR PID feedback            |
| AER_DO     | Aerobic basin (dedicated) | DO        | 2.0 mg/L setpoint         | Not on scan loop — continuous probe            |

s::can package: spectro::lyser V3 (UV-Vis BOD, COD, NO<sub>f</sub>, NH<sub>f</sub>, TSS, SAC254), pH::lyser in flow cell, ORP at anoxic-phase dwells, con::sole controller, sam::spector 5-port sequencer, P-SCAN recirculation pump with Y-strainer and 10 L buffer. Sample return to source. Dwell 8 min/port; full cycle ~40 min. Calibrate BOD/COD vs lab grab monthly; derive INF\_BOD\_TKN in MooreVIEW ST. Modbus TCP to edge historian at 1-min scan during active port.

7c. s::can Sequencer Port Schedule

| Port | Valve  | Phase          | Optical Parameters             | Notes                                          |
|------|--------|----------------|--------------------------------|------------------------------------------------|
| 1    | SV-101 | Influent       | BOD, COD, TSS, NH3, pH, SAC254 | Carbon load · C:N ratio · shock detection      |
| 2    | SV-102 | Pre-anoxic     | NO3, ORP, pH, TSS, COD         | Denitrification · MLSS · residual carbon       |
| 3    | SV-103 | IR stream      | NO3, pH                        | IR trim · nitrate in recycle                   |
| 4    | SV-104 | Aerobic outlet | NH3, pH, TSS                   | Nitrification · plus dedicated AER_DO in basin |
| 5    | SV-105 | Final effluent | BOD, NO3, NH3, TSS, pH         | Permit compliance · IR PID feedback            |

7d. AI-Based Process Optimization

The s::can multivariate data stream enables supervised ML that predicts optimal IR ratio, WAS rate, and DO setpoint. INF\_BOD and INF\_COD quantify carbon available for denitrification; when EFF\_NO3 is high but INF\_BOD/TKN is low, the model sets AI\_CARBON\_LIM rather than increasing IR. Phase pH and anoxic ORP provide redox context at each MLE step.

| AI Input Tag / Feature      | Role in Model                                        |
|-----------------------------|------------------------------------------------------|
| INF_BOD                     | Influent carbon load; C:N vs TKN; carbon-limit alarm |
| INF_COD                     | rbCOD proxy; supplemental carbon decision            |
| INF_BOD_TKN                 | Derived ratio — denitrification potential            |
| INF_TSS                     | Solids shock; normalize MLSS                         |
| INF_NH3                     | Nitrogen load forecast                               |
| INF_PH                      | pH inhibition / nitrifier stress                     |
| ANOX_NO3 + ANOX_ORP         | Denitrification state (single train)                 |
| T1_ANOX_* / T2_ANOX_*       | Per-train denitrification (dual layout)              |
| ANOX_COD                    | Residual carbon in anoxic — carbon limited?          |
| IR_NO3                      | IR effectiveness                                     |
| AER_NH3 + AER_DO            | Nitrification health                                 |
| EFF_NO3 + EFF_NH3 + EFF_BOD | Permit feedback · model reward                       |
| Phase pH (all ports)        | Process stability across MLE phases                  |

Recommended AI outputs (constrained setpoints written to MooreVIEW tags):

- AI\_IR\_SP — suggested IR ratio (bounded 2.5–4.5× Q); suppressed when AI\_CARBON\_LIM = 1.
- AI\_WAS\_SP — suggested WAS mL/min from ANOX\_TSS trend vs target MLSS band.

- AI\_DO\_SP — suggested aerobic DO mg/L when AER\_NH3 rising and ANOX\_TSS stable.
- AI\_CARBON\_LIM — flag when INF\_BOD/TKN < 3.5 or ANOX\_COD residual high with ANOX\_NO3 > 8 mg/L.
- AI\_ALARM — anomaly score when INF\_TSS or phase pH deviates >2<2 g&öÖ rÖF y baseline.

Deployment path: (1) collect 90 days historian data with rule-based IR trim; (2) train regression/gradient model offline on NH f/NO f/TSS features; (3) run inference on edge PC alongside MooreVIEW ST program; (4) validate AI vs manual operation for 30 days before permit reliance. AI never bypasses hard limits on EFF\_NO3 >10 mg/L or EFF\_NH3 >2 mg/L — those trigger immediate rule-based IR increase and operator alert.

## 8. Reactor Sizing

| Zone                  | Volume    | % of Total | Function                                  |
|-----------------------|-----------|------------|-------------------------------------------|
| Pre-anoxic            | 1,400 gal | 35%        | Denitrification; mixer only, no aeration  |
| Aerobic IFAS          | 2,600 gal | 65%        | Nitrification; fixed PVA + CRS sidestream |
| Settling (integrated) | 400 gal   | —          | Solids separation, RAS/WAS                |
| Total biological      | 4,000 gal | 100%       | HRT "H 2 days @ avg flow                  |

PVA foam media (fixed in SS cages, 5 mm max slot) — total fill ~170 ft³:

- Fill fraction: 50% of aerobic volume (~1,300 gal displaced)
- Media type: PVA foam cubes/blocks, 10–20 mm
- Surface area: 400–600 m²/m³ (vendor-specific)
- Effective biofilm SRT: 30–60+ days on carriers

## 9. CRS Nanobubble Aeration System

Equipment: CRS Nanobubble Generator (NBG) by Chemical Reduction Systems (chemicalreduction.com). The CRS NBG is a passive inline/sidestream device using hydrodynamic cavitation and ionization of entrained gases. It requires no electricity, no external gas input, and has no moving parts. Housing: 316 or 304 stainless steel; core alloy suitable for NSF/ANSI 61 applications. Factory warranty: 5 years; design service life: 10+ years.

| Component            | Specification                       |
|----------------------|-------------------------------------|
| CRS NBG model        | 1" line, sidestream per train       |
| Sidestream flow      | 10-15 continuous                    |
| Sidestream pump      | 0.33 HP, 24/7 duty                  |
| Aerobic DO setpoint  | 2.0 mg/L                            |
| Backup aeration      | 0.15 HP fine-bubble (DO trim)       |
| Anoxic mixing        | 0.25 HP submersible mixer per train |
| Estimated O , demand | 0.8-1.2 lb O ,/d                    |

## 10. Recycle & Solids Management

| Stream                | Rate                        | Equipment                   |
|-----------------------|-----------------------------|-----------------------------|
| Internal recycle (IR) | 3.5× Q = 7,000 GPD (~5 gpm) | VFD centrifugal per train   |
| RAS                   | 75% Q = 1,500 GPD (~1 gpm)  | Return to anoxic            |
| WAS                   | 5-10 gal/d                  | Timed valve; SRT 12–20 d    |
| MLSS target           | 2,000-3,000 mg/L            | Suspended + biofilm biomass |

IR ratio is trimmed on effluent NO *f*-N: increase IR when NO *f*-N >10 mg/L (max 4.5×); decrease when NO *f*-N <6 mg/L (min 2.5×).

## 11. Expected Performance

| Parameter           | Expected Effluent        |
|---------------------|--------------------------|
| BOD ...             | 5–8 mg/L                 |
| TSS                 | 5–10 mg/L                |
| NH <i>f</i> -N      | <1 mg/L                  |
| NO <i>f</i> -N      | 6–10 mg/L (with IR trim) |
| Total nitrogen (TN) | 10–14 mg/L               |

Nitrogen balance: ~35–40 mg/L NO *f*-N formed after full nitrification must be reduced ~75–80% via MLE denitrification to achieve <10 mg/L NO *f*-N. At BOD 200 mg/L and TKN 40 mg/L, influent rbCOD (~100–120 mg/L) is adequate without supplemental carbon under normal conditions.

## 12. Equipment Schedule

| Item                                | Qty   | Size / Model                               |
|-------------------------------------|-------|--------------------------------------------|
| FRP/concrete package tank           | 1     | 4,000 gal                                  |
| PVA foam + retention cages          | 1 lot | ~170 ft³; 50% aerobic fill                 |
| CRS NBG sidestream unit             | 1     | 1" SS; RFQ from CRS                        |
| Sidestream recirculation pump       | 1     | 10-15; 0.33 HP                             |
| Backup fine-bubble blower           | 1     | 0.15 HP with DO control                    |
| Anoxic submersible mixer            | 1     | 0.25 HP                                    |
| IR pump (VFD)                       | 1     | 5 gpm @ 8 ft TDH                           |
| RAS pump                            | 1     | 1 gpm                                      |
| Lamella / tube settler module       | 1     | 400 gal equivalent                         |
| scan recirculating analyzer package | 1     | scan · 5-port loop · BOD/pH/ORP            |
| AER_DO probe (dedicated)            | 1     | Continuous · blower PID — not on scan loop |
| MooreVIEW edge SCADA + AI module    | 1     | Project: mle-wastewater                    |

s::can installed budget ~\$55k (included in Section 16 CAPEX). Monthly lab verification of spectro::lyser BOD/ COD vs Standard Methods required for permit-grade reporting.

### 13. Instrumentation & Control

| Tag                   | Sample Port               | Function                                       |
|-----------------------|---------------------------|------------------------------------------------|
| INF_BOD               | Influent (SV-101)         | Carbon load · C:N ratio · shock detection      |
| INF_COD               | Influent (SV-101)         | Carbon load · C:N ratio · shock detection      |
| INF_TSS               | Influent (SV-101)         | Carbon load · C:N ratio · shock detection      |
| INF_NH3               | Influent (SV-101)         | Carbon load · C:N ratio · shock detection      |
| INF_PH                | Influent (SV-101)         | Carbon load · C:N ratio · shock detection      |
| INF_SAC254            | Influent (SV-101)         | Carbon load · C:N ratio · shock detection      |
| ANOX_NO3              | Pre-anoxic (SV-102)       | Denitrification · MLSS · residual carbon       |
| ANOX_ORP              | Pre-anoxic (SV-102)       | Denitrification · MLSS · residual carbon       |
| ANOX_PH               | Pre-anoxic (SV-102)       | Denitrification · MLSS · residual carbon       |
| ANOX_TSS              | Pre-anoxic (SV-102)       | Denitrification · MLSS · residual carbon       |
| ANOX_COD              | Pre-anoxic (SV-102)       | Denitrification · MLSS · residual carbon       |
| IR_NO3                | IR stream (SV-103)        | IR trim · nitrate in recycle                   |
| IR_PH                 | IR stream (SV-103)        | IR trim · nitrate in recycle                   |
| AER_NH3               | Aerobic outlet (SV-104)   | Nitrification · plus dedicated AER_DO in basin |
| AER_PH                | Aerobic outlet (SV-104)   | Nitrification · plus dedicated AER_DO in basin |
| AER_TSS               | Aerobic outlet (SV-104)   | Nitrification · plus dedicated AER_DO in basin |
| EFF_BOD               | Final effluent (SV-105)   | Permit compliance · IR PID feedback            |
| EFF_NO3               | Final effluent (SV-105)   | Permit compliance · IR PID feedback            |
| EFF_NH3               | Final effluent (SV-105)   | Permit compliance · IR PID feedback            |
| EFF_TSS               | Final effluent (SV-105)   | Permit compliance · IR PID feedback            |
| EFF_PH                | Final effluent (SV-105)   | Permit compliance · IR PID feedback            |
| AER_DO                | Aerobic basin (dedicated) | Not on scan loop — continuous probe            |
| SCAN_ACTIVE_PORT      | Sequencer                 | Current port 1–N                               |
| SCAN_RUN / SCAN_FAULT | P-SCAN loop               | Pump status / system fault                     |
| AER_DO                | Aerobic basin             | Continuous DO 2.0 mg/L; blower PID             |
| AI_IR_SP              | AI Optimizer out          | Suggested IR setpoint (Auto mode)              |
| AI_WAS_SP             | AI Optimizer out          | Suggested WAS rate from TSS trend              |
| AI_CARBON_LIM         | AI Optimizer out          | Carbon-limited denitrification flag            |
| IR_FLOW / IR_RATIO    | IR pump                   | Flow and ratio monitoring                      |
| TANK_LVL              | Reactor                   | Level alarm                                    |

|               |                 |                          |
|---------------|-----------------|--------------------------|
| NANO_LOOP_RUN | Sidestream pump | CRS loop runtime / fault |
|---------------|-----------------|--------------------------|

Control logic summary:

- Sidestream CRS loop: run continuous (24/7).
- s::can sequencer: run continuous; alarm SCAN\_FAULT; log all ports each cycle.
- IF EFF\_NO3 > 10 mg/L AND NOT AI\_CARBON\_LIM !' increase IR +0.2×/hr (max 4.5×).
- IF AI\_CARBON\_LIM = 1 !' hold IR; alarm operator; evaluate supplemental carbon.
- IF EFF\_NO3 < 6 mg/L !' decrease IR " 0.2×/hr (min 2.5×).
- IF AER\_NH3 > 2 mg/L OR EFF\_NH3 > 1 mg/L !' increase DO or reduce WAS.
- IF ANOX\_TSS > 4,000 mg/L for 4 hr !' increase WAS per AI\_WAS\_SP; check settler.
- IF INF\_BOD/TKN < 3.5 (derived tag) !' set AI\_CARBON\_LIM; do not raise IR.
- IF ANOX\_ORP > +100 mV at anoxic port dwell !' alarm: denitrification lost.
- IF phase pH < 6.2 or > 8.8 at any port !' process pH alarm.

## 14. Tank Layout (Section)

```
+--- ANOXIC 1,400 gal -----+--- AEROBIC 2,600 gal + PVA ---+
| Influent + RAS -> Mixer | Fixed PVA foam cages (50% fill)|
| <- IR from aerobic      | CRS sidestream + backup diff. |
| ORP probe               | DO probe                       |
+-----+-----+-----> Settler -> Eff
```

## 15. Startup & Commissioning

- Fill reactor; seed with nitrifying return sludge or commercial culture.
- Install and cage PVA media permanently — floating media causes clarifier carryover.
- Commission anoxic mixer and IR/RAS loops before enabling CRS sidestream.
- Start CRS sidestream at 50% pump speed; ramp while monitoring AER\_DO and ANOX\_ORP.
- Allow 2–3 weeks biofilm growth on PVA before enforcing 10 mg/L NO *f*-N limit.
- Tune IR trim after 30 days of stable operation and seasonal temperature data.

## 16. Capital Cost Estimate (CAPEX)

Planning-level installed costs, USD, 2025–2026. ±25–35% until vendor quotes received.

| Category                                | Budget | Mid   | High  |
|-----------------------------------------|--------|-------|-------|
| MLE reactor tanks (1×125K or 4,000 gal) | \$26k  | \$44k | \$78k |
| Fixed PVA foam + cages                  | \$7k   | \$16k | \$31k |
| CRS sidestream aeration (all trains)    | \$8k   | \$14k | \$28k |
| Pumps, mixers, clarifiers               | \$16k  | \$30k | \$56k |
| Instrumentation + AI                    | \$10k  | \$24k | \$47k |
| Controls / SCADA                        | \$7k   | \$12k | \$25k |
| Electrical + piping + site              | \$20k  | \$36k | \$68k |
| Install + commissioning                 | \$13k  | \$24k | \$43k |

|                            |        |        |        |
|----------------------------|--------|--------|--------|
| Engineering + FDEP permits | \$10k  | \$20k  | \$37k  |
| TOTAL INSTALLED            | \$130k | \$200k | \$310k |

## 17. Project Budget Summary

| Line Item                                | Amount     |
|------------------------------------------|------------|
| Construction (mid installed)             | ~\$200,000 |
| Owner contingency (15%)                  | ~\$30,000  |
| Collection / EQ / site civil (allowance) | ~\$30,000  |
| First-year OPEX reserve                  | ~\$4,000   |
| TOTAL PROJECT BUDGET                     | ~\$275,000 |

Excluded from above: land acquisition, long collection systems, enclosed buildings, effluent UV/chlorination, and septage receiving.

## 18. Operating Cost Estimate (OPEX)

| Item                           | Annual Cost       |
|--------------------------------|-------------------|
| Power (CRS loop + trim blower) | \$120 – \$180     |
| Lab / compliance sampling      | \$800 – \$2,500   |
| Analyzer consumables           | \$0 – \$1,500     |
| WAS hauling                    | \$200 – \$600     |
| PVA media replacement reserve  | \$300 – \$800     |
| Maintenance contract           | \$500 – \$2,000   |
| Contract O&M visits            | \$1,000 – \$3,000 |
| TOTAL ANNUAL OPEX              | \$6,000-9,000     |

## 19. CRS Vendor RFQ Template

To: Chemical Reduction Systems (CRS) – [chemicalreduction.com](http://chemicalreduction.com)  
Subject: RFQ – CRS NBG for 2,000 GPD MLE-IFAS WWTP (Decentralized package site)

Application: 2,000 GPD MLE-IFAS, 4,000 gal Package Reactor, fixed PVA biofilm carriers, effluent target  $\leq 10$  mg/L NO<sub>3</sub>-N.

Request: 1× CRS NBG sidestream unit(s), 1" line, 316 SS, 10-15 continuous recirculation on mixed liquor per train.

Please quote: unit price, min/max flow, pressure drop, lead time, warranty, WWTP reference installations, and recommendation for backup aeration at 10°C.

## 20. Permitting & Design Notes

- Confirm whether permit limit is NO<sub>3</sub>-N (as nitrogen) or NO<sub>3</sub> as ion (stricter).
- If limit is NO<sub>3</sub> ion at 10 mg/L (~2.3 mg/L as N), add post-denitrification filter.

- s::can recirculating optical loop on all MLE phases — calibrate BOD/COD vs lab monthly.

## 21. Document Revision

| Rev | Date       | Description                        |
|-----|------------|------------------------------------|
| 0   | 2026-07-10 | Initial preliminary design package |